Science-Bitch Field Guide

WAFT Research Team

1. Science-Bitch Field Guide

Version: 1.0
Date: 2026-01-14

1.1 Overview

Science-Bitch is a command-line tool that implements the full scientific method workflow for hypothesis testing, experimentation, and evidence-based conclusions.

1.2 Quick Start

1.3 Scientific Method Workflow

1. Form Hypothesis

Create a testable hypothesis with:
- **Statement**: What you're testing
- **Prediction**: Expected outcome
- **Variables**: Independent, dependent, and control variables

2. Design Experiment

Design an experiment that:
- Tests your hypothesis
- Controls for confounding variables
- Collects measurable data

3. Capture Initial State (A)

Before running the experiment:
- Capture system state
- Record baseline measurements
- Document initial conditions

4. Run Experiment

Execute the experiment:
- Use controlled variables
- Collect data during execution
- Record all measurements

5. Collect Data (C)

During the experiment:
- Record all measurements
- Track dependent variables
- Note any observations

6. Capture Final State (B)

After the experiment:
- Capture final system state
- Compare with initial state
- Identify changes

7. Analyze Results

Analyze collected data:
- Verify or refute hypothesis
- Calculate confidence
- Draw conclusions

8. Generate Report

Create documentation:
- Experiment report
- Analysis results
- Recommendations

1.4 Command Options

Run full interactive workflow:
1. Form hypothesis interactively
2. Design experiment
3. Run experiment
4. Analyze results
5. Generate report

1.5 File Structure

1.6 Integration

- **Scientific Method Tool**: Core functionality from
- **PDF Generation**: Uses
- **Work Effort**: Tracked in
- **State Capture**: Automatic state snapshots
- **Data Collection**: Comprehensive data recording

1.7 Best Practices

1. **Clear Hypotheses**: Make hypotheses specific and testable
2. **Control Variables**: Keep control variables constant
3. **Multiple Trials**: Run multiple experiments for reliability
4. **Document Everything**: Record all observations
5. **Analyze Thoroughly**: Don't skip the analysis step

1.8 Troubleshooting

Command not found: Ensure you're in the project root and dependencies are installed.

Import errors: Run

1.9 Examples

See for complete examples.

For more information, see the work effort: WE-260112-az3z

Generate this field guide as PDF.

Generate project status report PDF.

PDF generation fails: Check that WeasyPrint is installed and working.
